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The house behind VELORA.

L'ORéAL LUXE DIVISION

At L'Oréal Paris, beauty is where science meets emotion. Velora Paris embodies this vision through intelligent haute perfumery. It is a premium adaptive fragrance platform transforming perfume from a static gesture into a living ritual. A couture three-cartridge design, intuitive touch interface, and precision diffusion work in harmony. Each release is shaped by advanced sensing and environmental awareness, delivering an evolving fragrance experience inspired by Parisian elegance and modern luxury.

Innovation, Designed for the Senses

◆ Touch as Activation

An intuitive, tactile interface that initiates an adaptive, data-driven fragrance experience

◆ Adaptive Intelligence

A system that reads the body and the environment to refine scent delivery in real time

◆ Art × Science × Technology

A new generation of perfumery — where emotion, engineering, and elegance converge

Why Different Fragrances Matter

Because scent doesn't just surround you — it speaks directly to your brain, shaping emotion, memory, and perception.

Figure 1 : Projected Growth of the Global Fragrance Market (2025–2035)

Consumer Problems

- Mood Misalignment: Conventional fragrances remain **static**, lacking the ability to **adapt to dynamic emotional states**
- Inconsistent Longevity: Fragrance performance varies due to **individual body chemistry and ambient conditions**
- Sensitivity Concerns: **Fixed formulations** may cause **skin irritation for sensitive users**
- Limited Personalization: Traditional scents offer **minimal alignment with individual preferences and moods**
- Restricted Application Control: Manual application provides **limited control over fragrance intensity and release timing**

(Tap the icon to view the references.)

Market Insights

- A Growing Luxury Category: Fragrance projected to grow from **$25.8B (2025)** to **$44.8B by 2034**
- Rise of Niche & Exclusivity: Niche fragrances expected to **double to $7.6B by 2032**, driven by demand for **uniqueness**
- Personalization as the New Luxury: Consumers increasingly seek **tailored fragrance experiences**
- Digital Discovery Culture: **Social media and e-commerce** reshaping fragrance exploration
- Conscious Luxury: Rising demand for **sustainable ingredients** and **refillable formats**

L'ORÉAL

Velora Paris - Intelligent fragrance responsive to emotion and environment.

- Smart Fragrance System: Delivers **real-time personalized scents**, responding to **emotional state and ambient conditions**
- Biometric Interaction: **Fingertip contact** captures **physiological signals** to detect **emotional state**
- Intelligent Activation: **Biometric range** triggers the **selected fragrance cartridge**
- Environmental Optimization: **Temperature and humidity** refine **diffusion intensity**
- Responsive Dispensing: Fragrance **releases during contact** and **stops upon release**

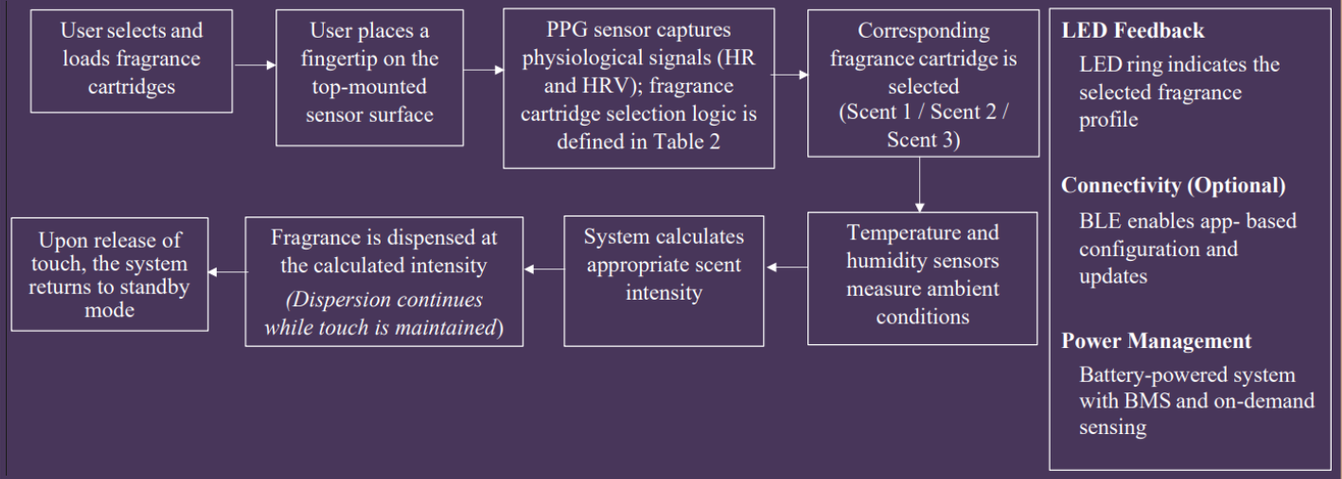


Figure 2: Functional workflow of the intelligent fragrance dispensing device

Table 1 : Mood Classification and Fragrance Response Mapping

Mood	HR Range (bpm)	HRV Level (ms RMSSD)	Triggers	LED Color	Cartridge Released
Energized / Anxious	>90	Low (<25)	High arousal, fight-or-flight	Red / Orange	Cartridge 1 (e.g., citrus / energizing)
Stressed / Tense	70–90	Low (<25)	Sympathetic dominance, rigid pulse	Blue / Purple	Cartridge 2 (e.g., lavender / woody calming)
Neutral / Balanced	60–80	High (>35)	Parasympathetic balance, flexible beats	Green / White	Cartridge 3 (e.g., floral / fresh daily)

Table 2 : Environmental Adaptation and Scent Diffusion Profiles

Profile	Environmental Condition	System Response	User Experience
Strong	Cold + Dry	Increased scent diffusion	Scent feels present & clear
Normal	Mild / Balanced conditions	Standard scent diffusion	Familiar, balanced fragrance
Soft	Hot + Humid	Reduced scent diffusion	Refined, close-to-skin scent

Table 3 : Biometric and Environmental Response Mapping

HR (bpm)	HRV (RMSSD)	Temperature	Humidity	System Response	Fragrance Output
95 bpm	18 ms	20°C	35%	Increased scent diffusion	Citrus / Energizing scent (Cartridge 1)
78 bpm	22 ms	25°C	50%	Standard scent diffusion	Lavender / Woody calming scent (Cartridge 2)
65 bpm	40 ms	32°C	75%	Reduced scent diffusion	Soft floral scent (Cartridge 3)

Sensor-Driven Prototype Demonstration

- Prototype integrates **Arduino Nano, PPG Heart-Rate Sensor, LED indicators, DHT11 Temperature–Humidity Sensor, and dual solenoid actuators**
- **First 3 BPM readings ignored**, **next 5 averaged** for **stable heart-rate detection**, with **temperature and humidity monitoring**
- **Activation logic: < 70 BPM → Yellow LED | ≥ 70 BPM → Green LED**
- **Solenoid-based actuation:** Green LED triggers **Solenoid 1**, while Yellow LED triggers **Solenoid 2**, demonstrating sensor-driven fragrance cartridge activation

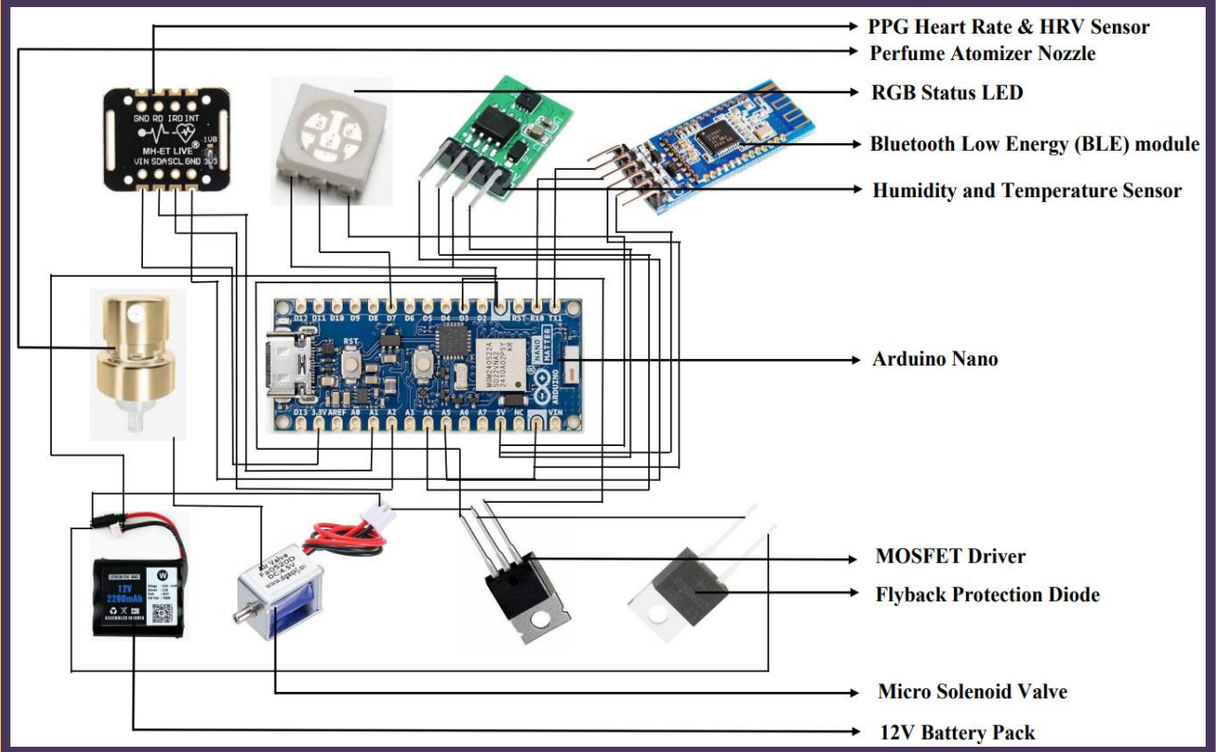


Figure 3 : Hardware architecture of the smart perfume diffuser

Table 4 : Bill of Materials (BOM) and Cost Analysis

S.No	Description	Quantity	Unit Cost (₹)	Total Cost (₹)
1	Arduino Nano V3	1	300	300
2	HM-10 BLE Bluetooth Module	1	280	280
3	MAX30101 PPG module	1	1,054	1,054
4	Humidity & Temperature Sensor (DHT11)	1	100	100
5	MOSFET (IRLZ44N)	2	32	64
6	Flyback Protection Diode (IN 4007)	2	0.85	1.7
7	Mini Solenoid Valve (12V 2A)	2	544	1,088
8	LEDs (Green, Yellow)	2	1	2
9	Adapter (12V 2A)	1	350	350
10	220Ω resistor	2	0.5	1
11	10kΩ resistor	2	0.5	1
12	Breadboard	1	60	60
13	Wires		150	150
12	3D Printed Parts		1,800	1,800
Total Cost			₹5,252	

Prototype Cost Insights

- **Total prototype cost:** ₹5,252/- , a low-cost proof-of-concept
- **Electronics (~60%):** Core sensing and control components
- **Modular design:** Supports future luxury upgrades and refillable cartridges

Figure 4: SolidWorks CAD Design of the Velora Paris Device

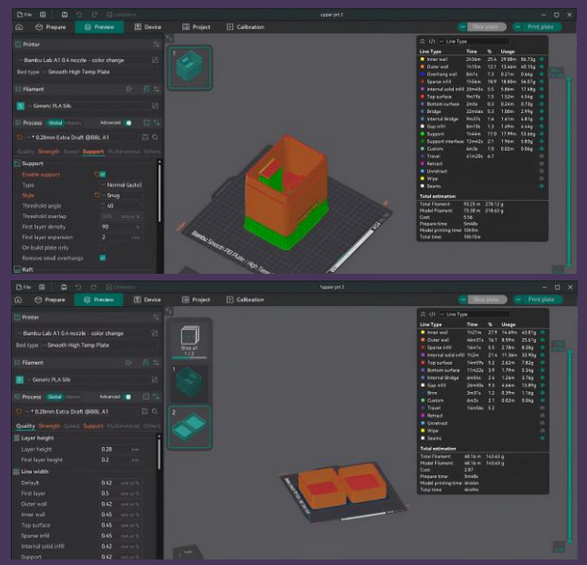


Figure 5: Bambu Studio Prototype Slicing

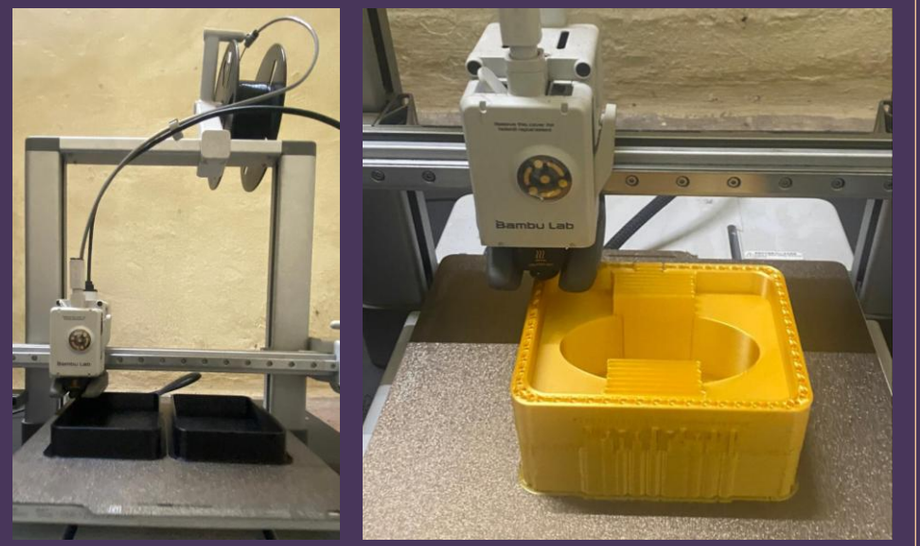


Figure 6: Bambu Lab 3D Printing of the Velora Paris Prototype

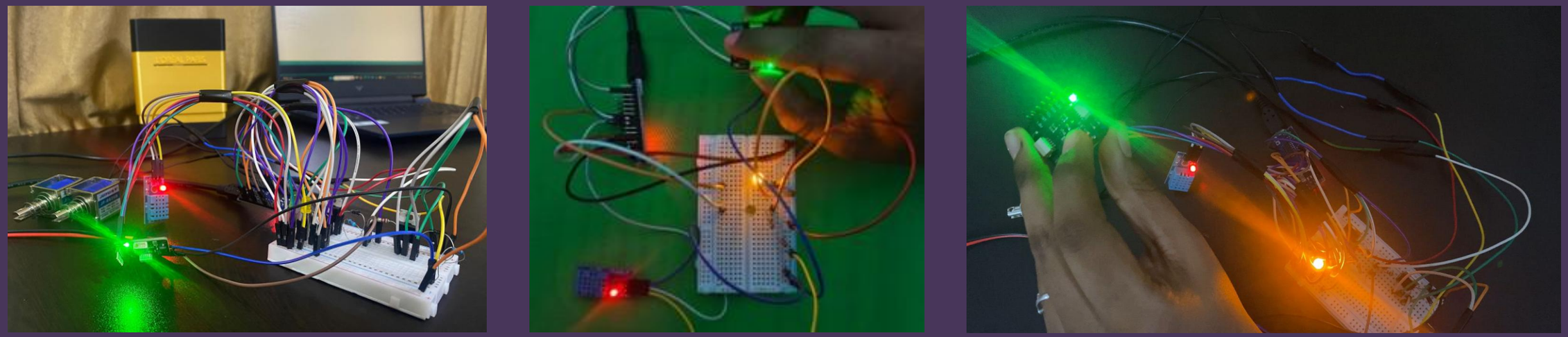


Figure 7: Smart Fragrance Prototype Assembly and Testing

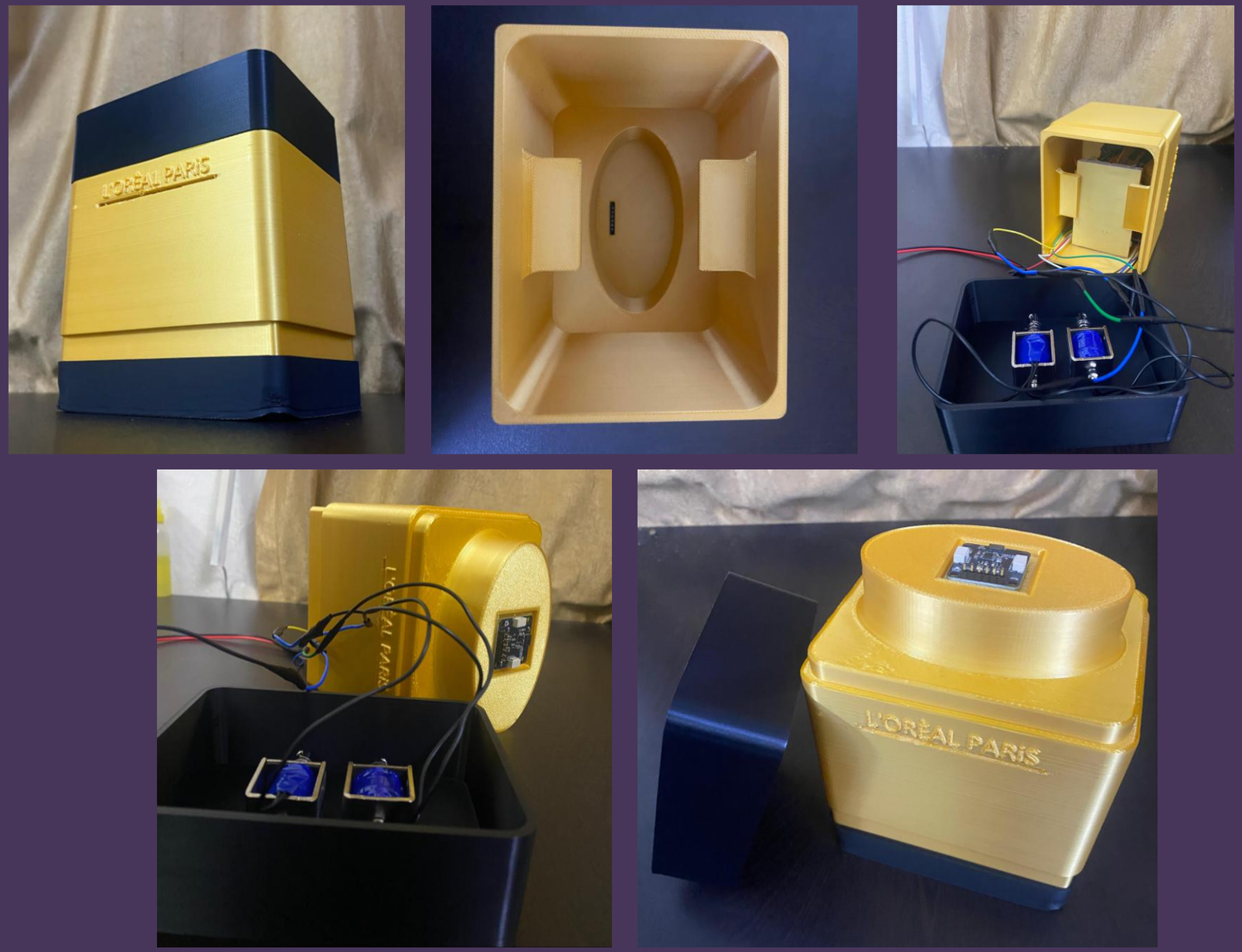
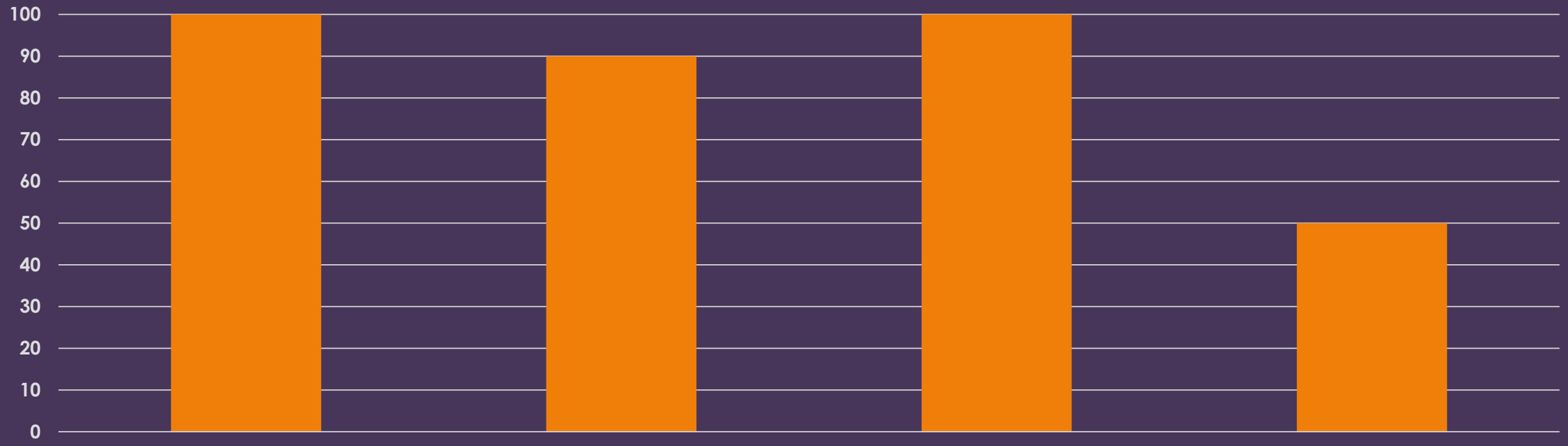


Figure 7: Precision Engineered Prototype Enclosure and Assembly

Prototype Development Progress



Phase 1: Concept & System Design

- Defined adaptive fragrance sensing architecture
- Developed HR/HRV-based mood detection logic

Phase 2: Functional Prototype

- Integrated sensors with Arduino Nano platform
- Verified individual sensor functionality and system response

Phase 3: Product Engineering

- Designed CAD housing for device architecture
- Integrated sensing and control modules for prototype output

Phase 4: Prototype Fabrication

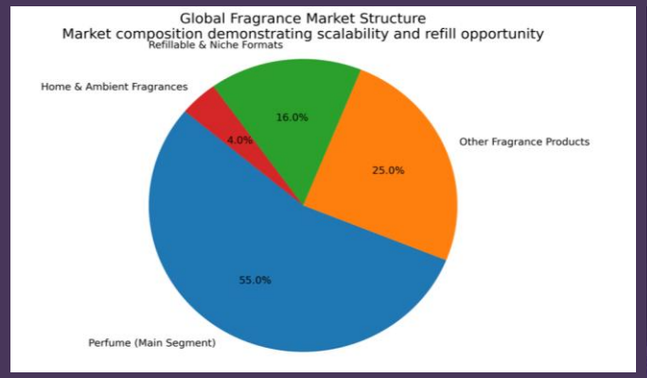
- 3D printed device housing
- Assembled electronics and finalized working prototype

GitHub Link: https://github.com/Anu-310/velora-paris-loreal/tree/main

Scalability & Business Viability

(Built for global L'Oréal Luxe deployment)

- Universal 3-Cartridge Platform: Works with any luxury fragrance
- Smart Upgrade for All Scents: Adds mood-sensing to existing perfumes
- Climate-Adaptive Diffusion: Auto-adjusts for humidity/temperature
- Revenue Model: Device + refills + app subscriptions



Key Success Indicators (KPIs)

(Brandstorm-aligned)

(Tap the Search References icon for instant, pure elegance.)

- Engagement: **78%** weekly active users, **3.2** scent switches/month, **65%** retention (vs industry **12.8%**)
- Business: **4.2** refills/device/year, **68%** cross-fragrance usage, **$185** LTV/device (3x industry average)
- Sustainability: **82%** bottle cycle reduction, **70%** refill adoption, **65%** material savings
- Brand: **76%** daily ritual integration, High emotional affinity score, **42%** advocacy rate

Sustainability & Responsible Luxury

(L'Oréal for the Future | Refill revolution)

- Long-life device + 3-cartridge refills: **82%** less packaging, **65%** material reduction
- Precision Mood-Triggered Spray: Optimized diffusion cuts fragrance waste by **40%**
- Refill-First Design: Compatible with all L'Oréal Luxe scents + existing refill stations
- Durable luxury object: One device replaces **15+** disposable bottles over 3 years

Consumer & Brand Impact

(L'Oréal Luxe | Intelligent Perfumery)

- Mood-Responsive Ritual: Perfume adapts instantly to your emotional state
- Daily Engagement: **3.2** scent switches/month via fingertip mood detection
- Orchestrated Fragrance Experience: Iconic scents turn smart & mood-responsive
- Premium Differentiation: Biosensor personalization no competitor offers